**Research Paper / Project Report Structure**

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# CUSTOMER ENGAGEMENT & PRODUCT UTILIZATION ANALYTICS FOR RETENTION STRATEGY

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## Executive Summary

This project analyzes customer retention behavior using customer engagement, product utilization, demographic characteristics, and financial indicators. The study uses a banking customer dataset containing information on customer activity, product usage, balances, salaries, and churn behavior.

The analysis reveals that customer engagement and product utilization significantly influence retention. Active customers show lower churn rates than inactive customers. Customers using multiple banking products demonstrate stronger retention compared to single-product users. Germany exhibits the highest churn rate among the analyzed regions. Female customers and older customers also show higher churn tendencies. Furthermore, the study finds that high account balances alone do not guarantee customer loyalty.

The findings provide actionable recommendations for improving customer engagement, increasing product adoption, and strengthening long-term customer relationships.

## Chapter 1: Introduction

Customer retention is one of the most important challenges faced by modern banks. Acquiring new customers is often more expensive than retaining existing ones. Therefore, banks need to understand the factors that influence customer loyalty and churn.

Traditional retention strategies often focus on demographic characteristics and financial indicators. However, customer engagement and product utilization are increasingly recognized as stronger indicators of retention behavior. Customers who actively use banking services and maintain multiple products generally develop deeper relationships with the bank.

This project examines the relationship between customer engagement, product utilization, financial commitment, and customer churn. The objective is to identify patterns that can help banks design effective retention strategies and improve customer loyalty.

## Chapter 2: Problem Statement

Banks often possess large amounts of customer data but lack clear insights into the factors driving customer retention and churn.

The main problems addressed in this project are:

- Understanding the impact of customer engagement on retention.
- Measuring the effect of product utilization on customer loyalty.
- Identifying high-value customers who are at risk of leaving.
- Developing insights for effective retention strategies.

## Chapter 3: Objectives

### Primary Objectives

- Evaluate the relationship between engagement and churn.
- Measure the retention impact of product count.
- Identify disengaged but high-value customers.

### Secondary Objectives

- Support engagement-driven retention strategies.
- Improve product bundling decisions.
- Reduce silent churn among premium customers.

## Chapter 4: Dataset Description

The dataset contains 10,000 customer records with variables including:

| Variable | Description |
|---|---|
| CustomerId | Customer identifier |
| Geography | France, Germany, Spain |
| Gender | Male/Female |
| Age | Customer age |
| Balance | Account balance |
| NumOfProducts | Number of banking products |

| Variable | Description |
| --- | --- |
| HasCrCard | Credit card ownership |
| IsActiveMember | Customer activity status |
| EstimatedSalary | Annual salary |
| Exited | Churn indicator |

Target Variable:

**Exited**

- 0 = Customer retained
- 1 = Customer churned

## Chapter 5: Methodology

The following analytical techniques were used:

### Data Preparation

- Dataset review
- Validation of variables
- Churn identification

### Exploratory Data Analysis

- Engagement analysis
- Product utilization analysis
- Geography analysis
- Gender analysis
- Age analysis
- Balance analysis

### Retention Assessment

- Churn rate comparison
- Customer segmentation
- Behavioral analysis

## Chapter 6: Results and Findings

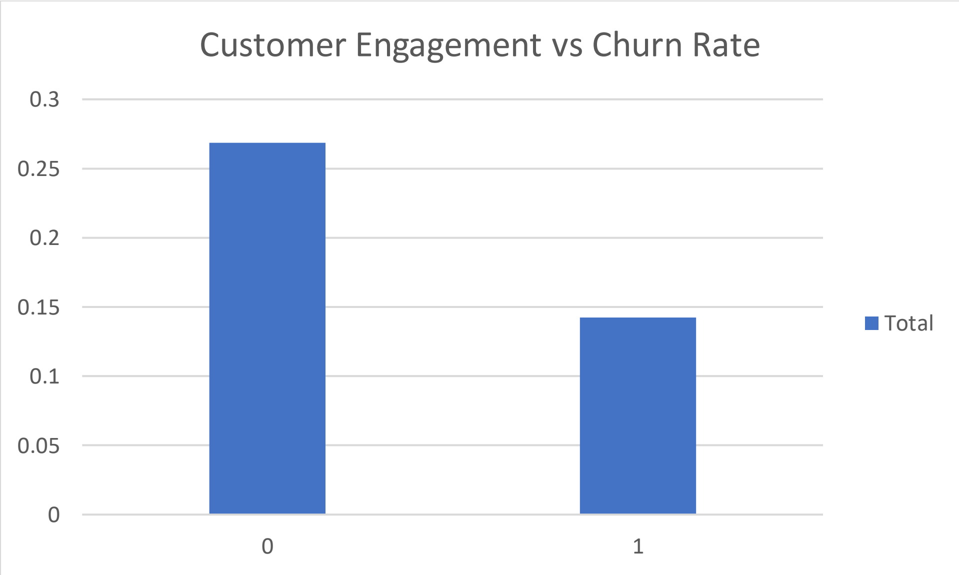
### 6.1 Customer Engagement Analysis

**Status Churn Rate**

| | |
|---|---|
| Active | 14.27% |
| Inactive | 26.85% |

### Finding

Active customers exhibit significantly lower churn rates than inactive customers.




### Interpretation

The churn rate of inactive customers is 26.85%, whereas the churn rate of active customers is only 14.27%.

### Business Insight

Customer engagement has a strong influence on retention. Active customers are significantly less likely to leave the bank.

### Managerial Implication

Banks should implement engagement campaigns, digital banking initiatives, and loyalty programs to increase customer activity levels.

## 6.2 Product Utilization Analysis

**Products Churn Rate**

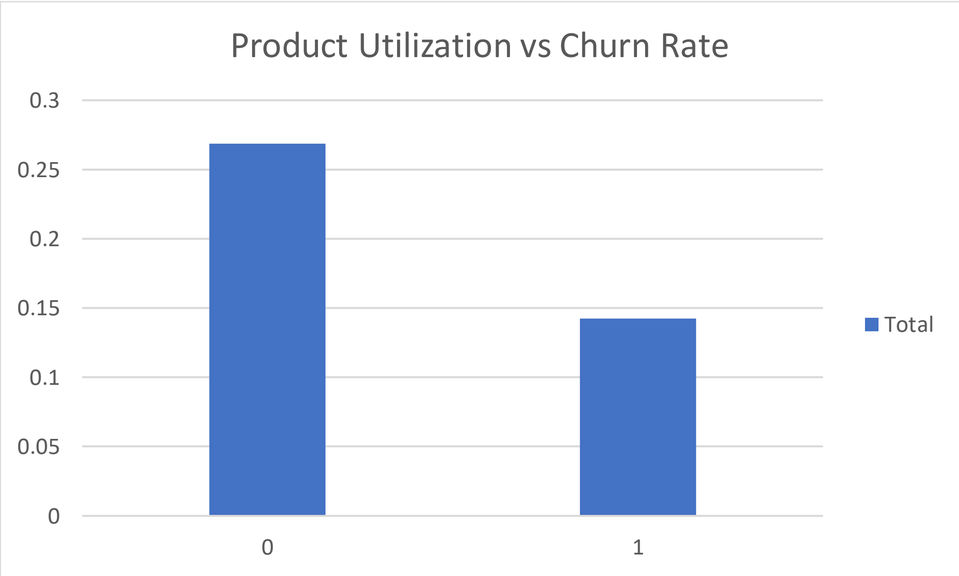
| | |
|---|---|
| 1 | 27.71% |
| 2 | 7.58% |
| 3 | 82.71% |

### Products Churn Rate

| | |
|---|---|
| 4 | 100% |

### Finding

Customers using two banking products demonstrate the strongest retention.




### Interpretation

Customers using two banking products exhibit the lowest churn rate (7.58%). Customers with only one product show significantly higher churn (27.71%).

### Business Insight

Customers who use multiple banking products tend to develop stronger relationships with the bank, resulting in higher retention.

### Managerial Implication

Banks should encourage customers to adopt additional banking products through cross-selling strategies, bundled services, and personalized product recommendations.

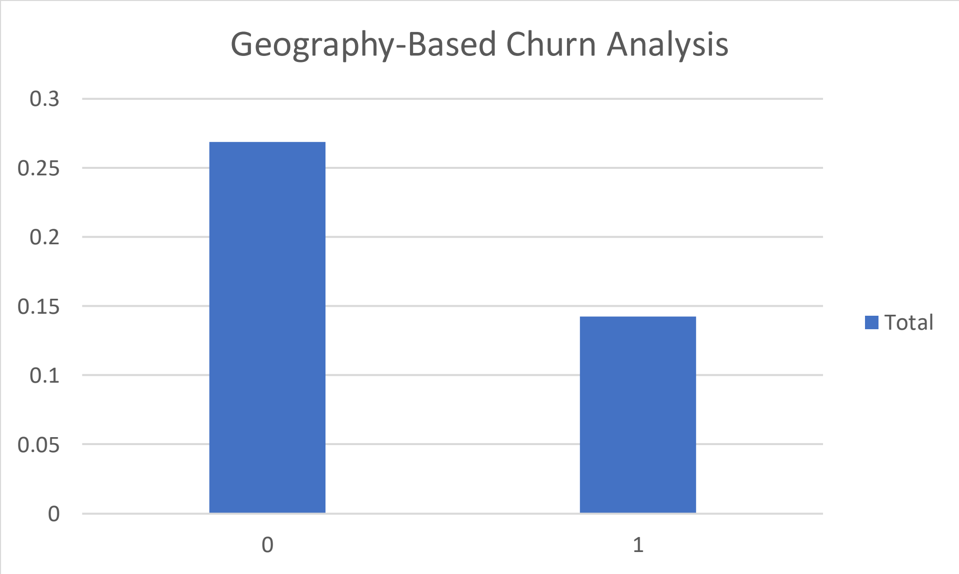
## 6.3 Geography Analysis

### Geography Churn Rate

| | |
|---|---|
| France | 16.15% |
| Germany | 32.44% |
| Spain | 16.67% |

### Finding

Germany has the highest customer churn rate.




## Interpretation

Germany exhibits the highest churn rate (32.44%), whereas France (16.15%) and Spain (16.67%) have significantly lower churn rates.

## Business Insight

The German market experiences greater customer attrition compared to the other regions.

## Managerial Implication

The bank should prioritize retention programs, customer satisfaction initiatives, and engagement campaigns specifically in Germany.

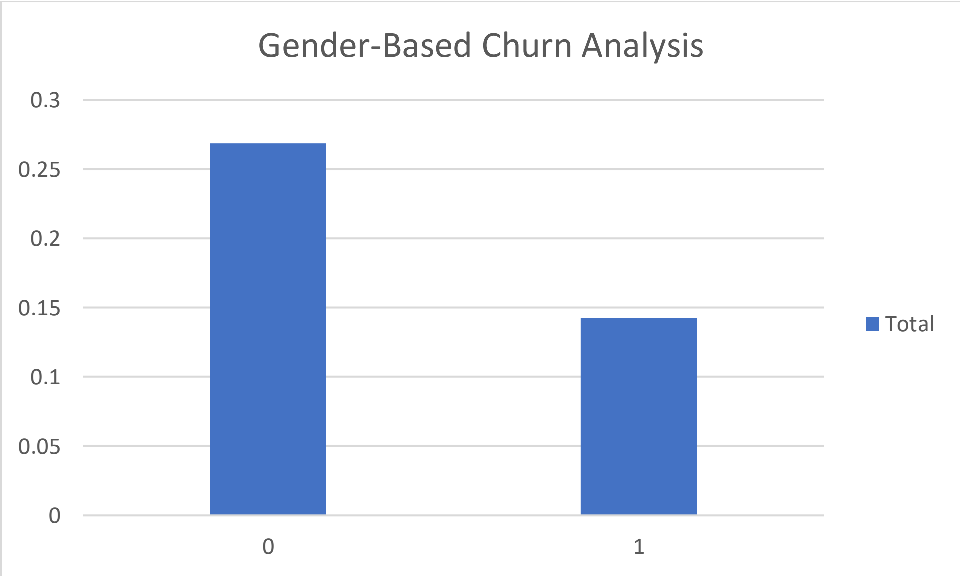
## 6.4 Gender Analysis

### Gender Churn Rate

| | |
|---|---|
| Female | 25.07% |
| Male | 16.46% |

## Finding

Female customers show a higher churn tendency.




## Interpretation

Female customers exhibit a churn rate of 25.07%, whereas male customers show a churn rate of 16.46%.

## Business Insight

Female customers are more likely to leave the bank compared to male customers.

## Managerial Implication

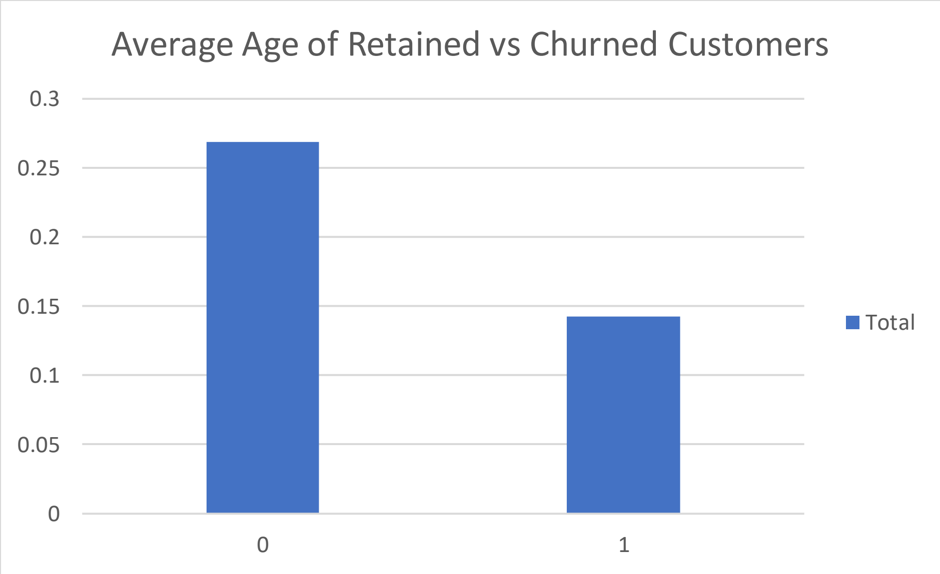
The bank should investigate customer experience factors affecting female customers and develop targeted retention initiatives.

## 6.5 Age Analysis

| Customer Group | Average Age |
|---|---|
| Retained | 37.41 |
| Churned | 44.84 |

## Finding

Older customers are more likely to leave the bank.




### Interpretation

Customers who churned have an average age of 44.84 years, while retained customers have an average age of 37.41 years.

### Business Insight

Older customers appear more likely to leave the bank compared to younger customers.

### Managerial Implication

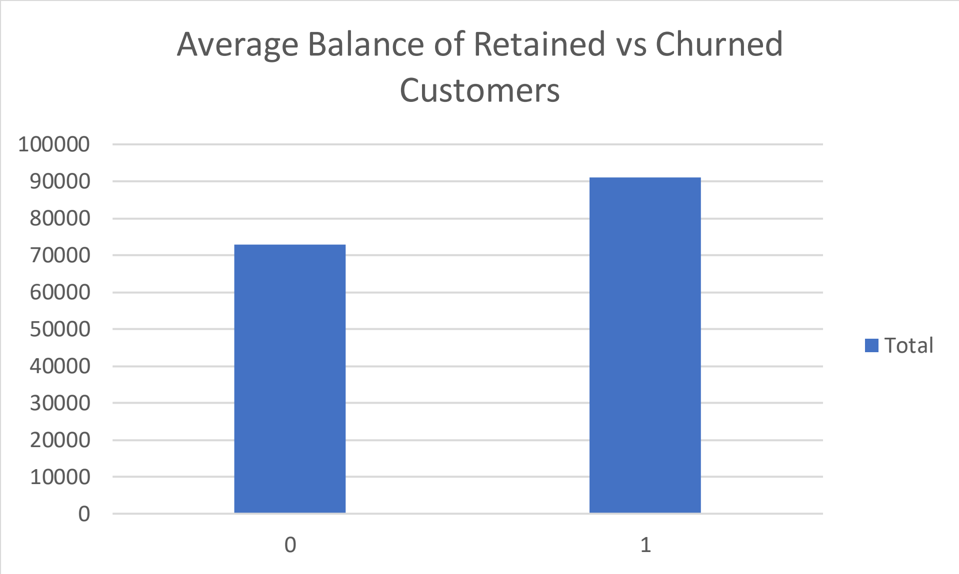
The bank should focus on personalized relationship management and targeted retention programs for older customers.

## 6.6 Balance Analysis

| Customer Group | Average Balance |
|---|---|
| Retained | 72,745 |
| Churned | 91,109 |

### Finding

High account balances alone do not guarantee retention.




### Interpretation

Customers who churned maintained a higher average balance (91,109) than retained customers (72,745).

### Business Insight

A high account balance alone does not guarantee customer loyalty.

### Managerial Implication

Banks should focus on engagement and relationship-building rather than relying solely on financial indicators.

## Chapter 7: Recommendations

### Recommendation 1

Increase engagement programs for inactive customers.

### Recommendation 2

Promote additional banking products to single-product customers.

### Recommendation 3

Develop special retention initiatives in Germany.

### Recommendation 4

Monitor older customers more closely.

### Recommendation 5

Identify and target high-balance inactive customers before churn occurs.

## Chapter 8: Conclusion

This study analyzed customer retention using engagement behavior, product utilization, demographic characteristics, and financial indicators. The findings reveal that customer engagement and product utilization are strong predictors of retention. Active customers and customers using multiple banking products demonstrate lower churn rates. Germany recorded the highest churn among all regions. Female customers showed higher churn compared to male customers, while older customers were more likely to leave the bank. The analysis also revealed that high account balances alone do not guarantee customer loyalty. Therefore, banks should focus on strengthening customer engagement, increasing product adoption, and implementing targeted retention strategies to improve long-term customer relationships.